

Activity: Remove Noise & Correct Distortion

Introduction

In digital imaging, noise and lens distortion can significantly impact image quality and accuracy. Noise, often caused by high ISO settings or environmental factors, reduces clarity and detail. Meanwhile, lens distortion, commonly found in wide-angle lenses, warps image geometry, leading to inaccuracies.

Learning Outcomes

1. **Noise Reduction** : Applying Gaussian and Median filters to remove noise while preserving image details.
 2. **Lens Distortion Correction** : Using OpenCV to correct geometric distortions and restore image accuracy.
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Working with Dev Container

To complete this activity in a reliable and fully configured environment, please refer to the instructions in the file: `README-devcontainer.md`. This guide walks you through opening the assignment in **Visual Studio Code** using a **Dev Container**, which automatically installs all necessary Python libraries and tools. Following that setup ensures that the notebook runs smoothly without manual configuration or missing dependencies. Make sure to open the **main activity folder** in VS Code and follow the steps outlined in the Dev Container guide before starting the notebook.

Instructions

Start by opening the Jupyter notebook provided in `Code/start` folder with this activity. This notebook is structured to help you learn **step by step**, combining explanations with code you can run and explore. Follow the tips below to get the most out of it. There will be tasks for you to complete as practice. These will be indicated by comments in the code block. In some cases, we will provide some starter code in a comment block. Other times we just provide a prompt for you to complete.

Working With Jupyter

With this type of environment, code can be executed out of order. That can have unintended consequences. Errors often arise from reusing variable names. If something does not appear to work correctly, try restarting this notebook and re-run all cells in order!

Conclusion

In this activity, you explored two fundamental image processing techniques that are critical for improving image quality and reliability. By reducing noise and correcting lens distortion, you learned how to enhance image clarity and restore geometric accuracy—both essential for effective computer vision analysis. These foundational skills prepare you to work with real-world image data and develop more robust vision systems.

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